

Project Report: Anime Recommendation System

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1 Introduction

In the contemporary digital era, the proliferation of online content has necessitated the development of sophisticated recommendation systems to assist users in navigating vast information spaces. Among these, the anime domain presents a unique challenge due to the diverse preferences of the user base and the hierarchical nature of anime metadata.

This project aims to design and implement a robust recommendation engine tailored for the anime dataset. The core objective is to move beyond simple popularity-based suggestions by leveraging Collaborative Filtering (CF) techniques. By analyzing user-item interaction patterns—specifically rating behaviors—we aim to uncover latent features that drive user preferences.

The primary contributions of this report are threefold:

1. **Data Exploration:** A comprehensive analysis of user-item interaction matrices to understand sparsity and interaction distributions.
2. **Noise Reduction:** Implementation of data preprocessing techniques to mitigate the "cold-start" problem and improve dataset quality.
3. **Model Evaluation:** Development and assessment of recommendation algorithms, evaluated against standard metrics such as Root Mean Square Error (RMSE) to ensure prediction accuracy.

Through this systematic approach, we intend to provide a scalable and effective solution capable of delivering personalized recommendations within the high-dimensional anime ecosystem.

2 Data Exploration and Matrix Analysis

To understand the dataset structure, we performed an initial analysis of the user-item interaction matrix. The following Python code was used to calculate the matrix density and sparsity:

Listing 1: Calculating Matrix Sparsity

```
1      # Calculate dimensions
2      num_users = rating_df['user_id'].nunique()
3      num_animes = rating_df['anime_id'].nunique()
4      total_ratings = len(rating_df)
5
6      # Sparsity calculation
7      density = total_ratings / (num_users * num_animes)
8      sparsity = 1 - density
9
10     print(f"Matrix Sparsity: {sparsity*100:.2f}%")
```

2.1 Visualizing the Long Tail

We visualized the interaction frequency to identify the "Long Tail" phenomenon, where a small number of anime titles receive a disproportionate number of ratings.

Listing 2: Plotting Interaction Frequency

```
1      # Plotting the Long Tail distribution
2      anime_popularity = rating_df.groupby('anime_id')['
      ↪ rating'].count()
3      plt.plot(np.sort(anime_popularity.values))
4      plt.yscale('log') # Log scale to handle the skewness
5      plt.title('Anime Interaction Frequency (Long Tail)')
6      plt.show()
```

2.2 Mathematical Context

The interaction dataset is modeled as a matrix $R \in \mathbb{R}^{U \times I}$, where U denotes the set of users, I denotes the set of anime titles, and r_{ui} represents the rating given by user u to anime i .

Given that most users interact with only a tiny fraction of the available anime, R is extremely sparse. The sparsity is mathematically defined as:

$$\text{Sparsity} = 1 - \frac{|\{r_{ui} \mid r_{ui} \neq 0\}|}{|U| \times |I|} \quad (1)$$

Furthermore, the interaction frequency visualized in the long-tail plot follows a power-law distribution. This is expressed as $P(x) \propto x^{-\alpha}$, indicating that a highly skewed minority of items receives the vast majority of interactions.

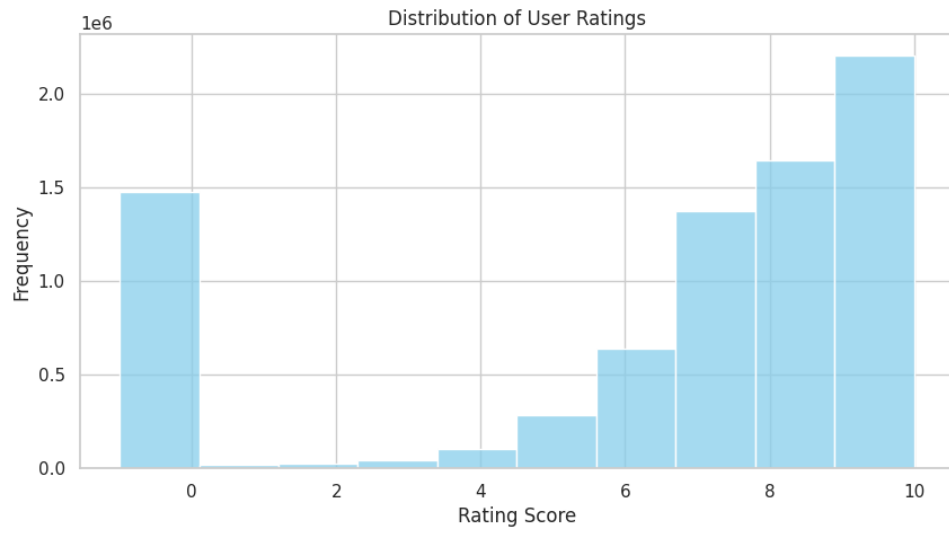


Figure 1: Distribution of User Ratings

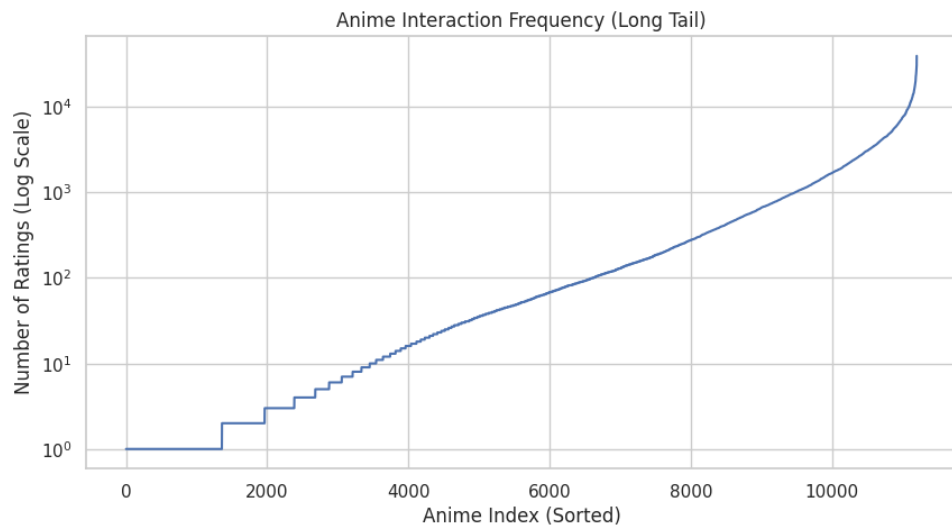


Figure 2: Long Tail Frequency (Logarithmic Scale)